

16th Feb 2026 / DAG360

⇒ GD in 1-dim.

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slope $\equiv f'$

- 1st order method $\equiv$ we make use of $f'(x_k)$
 - Bisection search $\Rightarrow$ use $f'(x_k)$ but cuts the space
 - here instead of cutting the space, we make small movements left or right on $f'(x_k)$
 - if $x_{k+1} = x_k - \alpha$ ($f'(x_k) > 0$)
if $f'(x_k) < 0$
 $x_{k+1} = x_k + \alpha$
- [slope is just a no. & not a line]

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$$x_{k+1} = x_k - \alpha \cdot \text{sign}(f'(x_k))$$

$$\begin{aligned} \text{sign}(n) &= 1, & n > 0 \\ &= -1, & n < 0 \\ &= 0, & n = 0 \end{aligned}$$

→ Signed Gradient descent

- if we run the above algo. on
 $\min. \frac{1}{2} x^2 \Rightarrow x_{k+1} = x_k - \alpha \cdot \text{sign}(x_k)$

if $\alpha = 2$ ($\alpha > 1$)
 $x_0 = 1$
 $x_1 = -1$
 $x_2 = +1$

} just toggling

states of convergence

for any constant α , it will not converge.
(true, because of non-smoothness)

Ex. Find Range of α s.t. do not toggle?

$x_0 \neq 0, \alpha > 0$

if $x_0 > 0, x_k = x_0 - k\alpha$ as it stays +ve.

so no toggling will happen iff. you hit 0 in finite steps

$$\exists N \in \mathbb{N} \text{ s.t. } x_0 - Nx = 0 \Leftrightarrow x = \frac{|x_0|}{N} \quad N \geq 1$$

$$x \in \left\{ \frac{|x_0|}{N} : N = 1, 2, 3, \dots \right\}$$

Ex. Pick $x_0 = 1, 1.5, -1.2, \dots$, $\alpha = 0.1, 0.01, 0.05$
 & find out how close you can get to $x^* = 0$.
 if α divides $|x_0|$ $\min_k |x_k| = 0$

$$\text{e.w.} \quad \min_k |x_k| = \min(a, a-a) \leq \frac{a}{2}$$

$$\frac{1}{0.1} \text{ (10 steps)}, \quad \frac{-1.2}{0.1} \text{ (12 steps)}$$

all the above pairs get exactly to $x^* = 0$.

- Constant doesn't work $\Rightarrow$ we have to diminish the step-size.

e.g. $\alpha = \frac{1}{k(1+k)^2}$ (for the sake of argument)

$$x_{k+1} = x_k - \frac{1}{(1+k)^2} (x_k)$$

$$x_0 = 100, \quad x_1 = 99, \quad x_2 = \dots \quad x^* = 0$$

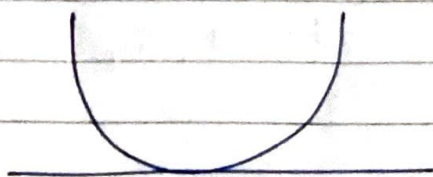
$$100 - \left(\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots \right) \quad \sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6}$$

$$\sum_{k \geq 1} \alpha_k = \infty \quad (\text{sum of entire learning rate must be } \infty)$$

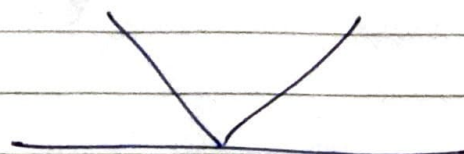
- signed GD ^{used} only left/right informaⁿ.
 It tells you the direcⁿ only.

ignores smoothness & does not exploit the smoothness property $\Rightarrow$ signed GD 1/1

$$f_1(x) = \frac{1}{2} x^2$$



$$f_2(x) = |x|$$



If we run signed GD on $f_1(x)$ & $f_2(x)$ starting at same $x_0 \neq 0$,

identical

$$\begin{cases} x_{k+1}^1 = x_k^1 - \alpha_k \text{sign} (f_1'(x_k)) \\ x_{k+1}^2 = x_k^2 - \alpha_k \text{sign} (f_2'(x_k)) \end{cases}$$

on $f_2(x)$ it should work better since, slope is constant.

(we don't know signal is not clear.)

$\Rightarrow$ Gradient descent

$$x_{k+1} = x_k - \alpha f'(x_k)$$

- now we might think that GD is faster on $f_2(x)$ than $f_1(x)$.

$$f_1(x) = \frac{1}{2} x^2$$

$$x_{k+1} = x_k - \alpha x_k \xrightarrow{\text{step-size}} = x_k (1 - \alpha) \xrightarrow{\text{slope}}$$

slope is steeper here

$$x_{k+1} = x_k - \alpha \text{sign}(x_k)$$

here it seems constant until it might start oscillating

- $f_1(x)$ has got better with GD. But this issue of $f_2(x)$ we still have the same problem of slope. $\rightarrow$ (Iterative averaging method)
 $\rightarrow$ because the learning rate is getting diminished by the slope itself.